Problem Statement – House Price Analysis You are a data analyst working with real estate data. Your task is to analyze the dataset to understand: Individual variable behavior (Univariate Analysis) Relationships between variables (Bivariate Analysis) Business insights from relationships (Interpretation)

```
import pandas as pd
df= pd.read_csv('/content/House Price India.csv')
df
```

	id	Date	number of bedrooms	number of bathrooms	living area	lot area	number of floors	waterfront present	number of views	condition of the house	...	Built Year	Renovation Year
0	6762810145	42491	5	2.50	3650	9050	2.0	0	4	5	...	1921	0
1	6762810635	42491	4	2.50	2920	4000	1.5	0	0	5	...	1909	0
2	6762810998	42491	5	2.75	2910	9480	1.5	0	0	3	...	1939	0
3	6762812605	42491	4	2.50	3310	42998	2.0	0	0	3	...	2001	0
4	6762812919	42491	3	2.00	2710	4500	1.5	0	0	4	...	1929	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
14615	6762830250	42734	2	1.50	1556	20000	1.0	0	0	4	...	1957	0
14616	6762830339	42734	3	2.00	1680	7000	1.5	0	0	4	...	1968	0
14617	6762830618	42734	2	1.00	1070	6120	1.0	0	0	3	...	1962	0
14618	6762830709	42734	4	1.00	1030	6621	1.0	0	0	4	...	1955	0
14619	6762831463	42734	3	1.00	900	4770	1.0	0	0	3	...	1969	2009

14620 rows × 23 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14620 entries, 0 to 14619
Data columns (total 23 columns):
 #   Column                                Non-Null Count  Dtype  
---  ------                                --------------  -----  
 0   id                                    14620 non-null  int64  
 1   Date                                  14620 non-null  int64  
 2   number of bedrooms                    14620 non-null  int64  
 3   number of bathrooms                   14620 non-null  float64
 4   living area                           14620 non-null  int64  
 5   lot area                              14620 non-null  int64  
 6   number of floors                      14620 non-null  float64
 7   waterfront present                    14620 non-null  int64  
 8   number of views                       14620 non-null  int64  
 9   condition of the house                14620 non-null  int64  
 10  grade of the house                    14620 non-null  int64  
 11  Area of the house(excluding basement) 14620 non-null  int64  
 12  Area of the basement                  14620 non-null  int64  
 13  Built Year                            14620 non-null  int64  
 14  Renovation Year                       14620 non-null  int64  
 15  Postal Code                           14620 non-null  int64  
 16  Lattitude                             14620 non-null  float64
 17  Longitude                             14620 non-null  float64
 18  living_area_renov                     14620 non-null  int64  
 19  lot_area_renov                        14620 non-null  int64  
 20  Number of schools nearby              14620 non-null  int64  
 21  Distance from the airport             14620 non-null  int64  
 22  Price                                 14620 non-null  int64  
dtypes: float64(4), int64(19)
memory usage: 2.6 MB
```

```
df.describe()
```

	id	Date	number of bedrooms	number of bathrooms	living area	lot area	number of floors	waterfront present	numbe v
count	1.462000e+04	14620.000000	14620.000000	14620.000000	14620.000000	1.462000e+04	14620.000000	14620.000000	14620.000
mean	6.762821e+09	42604.538646	3.379343	2.129583	2098.262996	1.509328e+04	1.502360	0.007661	0.233
std	6.237575e+03	67.347991	0.938719	0.769934	928.275721	3.791962e+04	0.540239	0.087193	0.766
min	6.762810e+09	42491.000000	1.000000	0.500000	370.000000	5.200000e+02	1.000000	0.000000	0.000
25%	6.762815e+09	42546.000000	3.000000	1.750000	1440.000000	5.010750e+03	1.000000	0.000000	0.000
50%	6.762821e+09	42600.000000	3.000000	2.250000	1930.000000	7.620000e+03	1.500000	0.000000	0.000
75%	6.762826e+09	42662.000000	4.000000	2.500000	2570.000000	1.080000e+04	2.000000	0.000000	0.000
max	6.762832e+09	42734.000000	33.000000	8.000000	13540.000000	1.074218e+06	3.500000	1.000000	4.000

8 rows × 23 columns

```
df.isna().sum()
```

	0
id	0
Date	0
number of bedrooms	0
number of bathrooms	0
living area	0
lot area	0
number of floors	0
waterfront present	0
number of views	0
condition of the house	0
grade of the house	0
Area of the house(excluding basement)	0
Area of the basement	0
Built Year	0
Renovation Year	0
Postal Code	0
Lattitude	0
Longitude	0
living_area_renov	0
lot_area_renov	0
Number of schools nearby	0
Distance from the airport	0
Price	0

dtype: int64

```
# drop null values or fill null values

df = df.fillna("0")
df = df.dropna()
```

```
df.isna().sum()
```

	0
id	0
Date	0
number of bedrooms	0
number of bathrooms	0
living area	0
lot area	0
number of floors	0
waterfront present	0
number of views	0
condition of the house	0
grade of the house	0
Area of the house(excluding basement)	0
Area of the basement	0
Built Year	0
Renovation Year	0
Postal Code	0
Lattitude	0
Longitude	0
living_area_renov	0
lot_area_renov	0
Number of schools nearby	0
Distance from the airport	0
Price	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

Q1. What is the average, median, and standard deviation of house prices?

```
df['Price'].mean(), df['Price'].median(), df['Price'].std()
```

```
(np.float64(538932.2183310534), 450000.0, 367532.3808039669)
```

Mean → overall average

Median → middle value (robust to outliers)

Std Dev → spread of prices

Q2. Which number of bedrooms is most common?

```
df['number of bedrooms'].mode()[0]
```

```
np.int64(3)
```

Q3. Check if the price data is skewed?

```
df['Price'].skew()
```

```
np.float64(4.269297720707116)
```

Positive skew → more low-priced houses, few very high-priced ones.

Negative skew → opposite.

Q4. What is the average price for each number of bedrooms?

```
df.groupby('number of bedrooms')['Price'].mean()
```

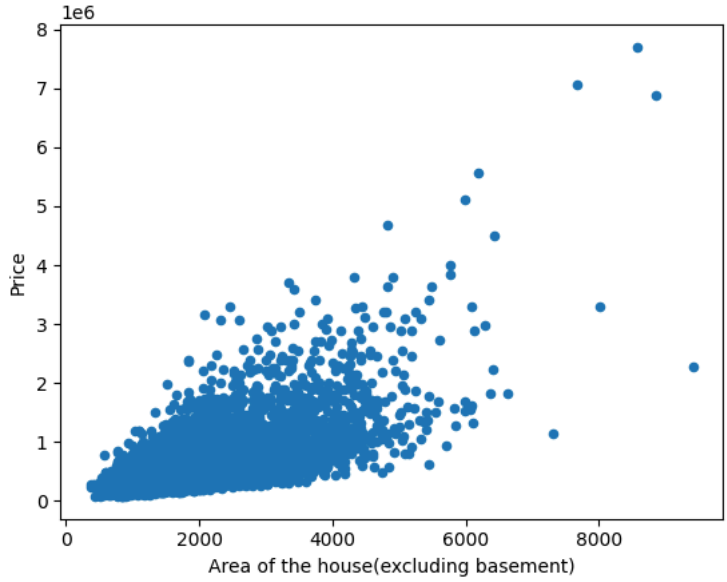
number of bedrooms	Price
1	3.089638e+05
2	3.985476e+05
3	4.632776e+05
4	6.361988e+05
5	7.752550e+05
6	8.375815e+05
7	1.016544e+06
8	1.208455e+06
9	7.766663e+05
10	8.200000e+05
11	5.200000e+05
33	6.400000e+05

dtype: float64

Q5. What is the relationship between living area and price? (Use : Area without Basement)

```
df.plot.scatter(x='Area of the house(excluding basement)', y='Price')
```

```
<Axes: xlabel='Area of the house(excluding basement)', ylabel='Price'>
```



Q6. Identify any anomalies where houses have high prices but low areas.

```
df[(df['Area of the house(excluding basement)'] < 1000) & (df['Price'] > df['Price'].quantile(0.9))]
```

	id	Date	number of bedrooms	number of bathrooms	living area	lot area	number of floors	waterfront present	number of views	condition of the house	...	Built Year	Renovation Year	Po
8517	6762811560	42621	3	1.75	1810	24586	1.0	0	4	4	...	1983	0	12

rows × 23 columns

Q7. Compare average price based on number of floors and houses with or without waterfront

```
df.groupby(['number of floors','waterfront present'])['Price'].mean()
```

number of floors	waterfront present	Price
1.0	0	4.323724e+05
	1	1.211081e+06
1.5	0	5.476162e+05
	1	1.232507e+06
2.0	0	6.357742e+05
	1	2.075931e+06
2.5	0	1.087654e+06
	1	3.280000e+06
3.0	0	5.558095e+05
	1	1.655000e+06
3.5	0	1.205875e+06

dtype: float64

Q8. Identify the minimum and maximum house price. What does this indicate?

```
df['Price'].min(), df['Price'].max()
```

```
(78000, 7700000)
```

Min → entry-level homes.

Max → luxury properties.

Indicates dataset covers wide market range.

Q9. Which location (zipcode/area) has the highest average price?

```
df.groupby('Postal Code')['Price'].mean().sort_values(ascending=False).head(1)
```

Postal Code	Price
122071	2.348311e+06

dtype: float64

Q10. Write at least 5 insights from your analysis.

1.House prices are positively skewed → few luxury homes inflate averages.

2.3-bedroom houses are most common.

3.Waterfront properties have significantly higher average prices.

4.Larger living area strongly correlates with higher price.

5.Certain zipcodes dominate premium pricing, showing location is a key driver